Frequency Analysis

ECSE 313 Group 16 Lab 3 Report

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Contents

1 Background **2**
1.1 Synthesis of Periodic Signals 2
1.2 Magnitude and Phase of Discrete-Time Systems 3

2 Continuous-Time Frequency Analysis **4**
2.1 Synthesis of Periodic Signals 4
2.2 Modulation Property 4
2.3 System Analysis 5

3 Discrete-Time Frequency Analysis **5**
3.1 Discrete-Time Fourier Transform 5
3.2 System Analysis 6

1 Background

1.1 Synthesis of Periodic Signals

Using the Fourier series expansion in the form

$$s(t) = a_0 + \sum_{k=1}^{\infty} A_k sin(2\pi k f_0 t + \theta_k) \tag{1}$$

for the two equations

$$s_1(t) = rect(t - \frac{1}{2}) \tag{2}$$

$$s_2(t) = rect(2t) - \frac{1}{2} \tag{3}$$

The piecewise function in Equation 2 is only 1 from $t = 0$ to $t = 1$ and 0 elsewhere. Therefore, the function becomes:

$$f_1(t) = \begin{cases} 1, & 0 < t < 1 \\ 0, & 1 < t < 2 \end{cases} \tag{4}$$

Using a period ($T_0 = 2$) from $t \in [0, 2]$ and $f_0 = \frac{1}{T} = \frac{1}{2}$, the a_0 value of Equation 2 becomes

$$a_0 = \frac{1}{T}\int_0^T f_1(t)dt = \frac{1}{2}\int_0^1 1dt = \frac{1}{2} \tag{5}$$

The fundamental frequency of the function is $w_0 = \frac{2\pi}{T} = \pi$. Moreover, a_k is

$$a_k = \frac{2}{T}\int_0^T f_1(t)cos(kw_0 t)dt = \int_0^1 cos(k\pi t)dt = \frac{sin(k\pi)}{k\pi} = 0 \tag{6}$$

a_k is zero because the sine function is zero for every integer value of k. The odd part of the signal, b_k, is defined as

$$b_k = \frac{2}{T}\int_0^T f_1(t)sin(kw_0 t)dt = \int_0^1 sin(k\pi t)dt = \frac{1 - cos(k\pi)}{k\pi} \tag{7}$$

In Equation 7, when k is even, $b_k = \frac{1-1}{k\pi} = 0$. When k is odd, $b_k = \frac{1-(-1)}{k\pi} = \frac{2}{k\pi}$. Moreover, $A_k = b_k$ and $\theta_k = 0$. Thus, Equation 2 becomes

$$s_1(t) = \frac{1}{2} + \sum_{k=1,3,5,...}^{\infty} \frac{2}{k\pi} sin(\pi k t) \tag{8}$$

The piecewise function in Equation 3 is only $1 - \frac{1}{2} = \frac{1}{2}$ from $t = -\frac{1}{4}$ to $t = \frac{1}{4}$ and $-\frac{1}{2}$ elsewhere. Therefore, the function becomes:

$$f_2(t) = \begin{cases} \frac{1}{2}, & |t| < \frac{1}{4} \\ -\frac{1}{2}, & \frac{1}{4} < |t| < \frac{1}{2} \end{cases} \tag{9}$$

Since $f_2(t)$ is an even function (symmetric across the y-axis), $b_k = 0$. Using a period ($T_0 = 1$) from $t \in [-\frac{1}{2}, \frac{1}{2}]$, the a_0 value of Equation 3 becomes

$$a_0 = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f_2(t)dt = \int_{-\frac{1}{2}}^{-\frac{1}{4}} -\frac{1}{2}dt + \int_{-\frac{1}{4}}^{\frac{1}{4}} \frac{1}{2}dt + \int_{\frac{1}{4}}^{\frac{1}{2}} -\frac{1}{2}dt = 0 \tag{10}$$

The fundamental frequency of the function is $w_0 = \frac{2\pi}{T} = 2\pi$ and $f_0 = \frac{1}{T} = 1$. Moreover, a_k is

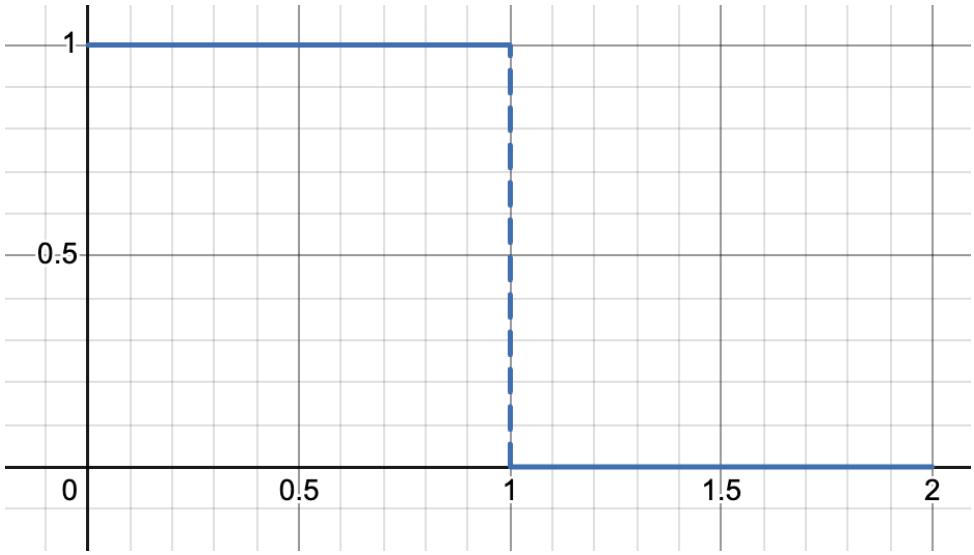

Figure 1: Sketch of Equation 2 from $[0, T_0]$

$$a_k = \frac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f_2(t)cos(kw_0t)dt = 2\int_{-\frac{1}{2}}^{-\frac{1}{4}} -\frac{1}{2}cos(2\pi kt) + 2\int_{-\frac{1}{4}}^{\frac{1}{4}} \frac{1}{2}cos(2\pi kt) + 2\int_{\frac{1}{4}}^{\frac{1}{2}} -\frac{1}{2}cos(2\pi kt) \quad (11)$$

$$a_k = -\left[\frac{sin(2\pi kt)}{2\pi k}\right]_{-\frac{1}{2}}^{-\frac{1}{4}} + \left[\frac{sin(2\pi kt)}{2\pi k}\right]_{-\frac{1}{4}}^{\frac{1}{4}} - \left[\frac{sin(2\pi kt)}{2\pi k}\right]_{\frac{1}{4}}^{\frac{1}{2}} \quad (12)$$

$$a_k = \frac{4sin(\frac{\pi k}{2}) - 2sin(\pi k)}{2\pi k} = \frac{2sin(\frac{k\pi}{2})}{k\pi} \quad (13)$$

$A_k = a_k$. When k is even, $A_k = 0$. For $k = 1, 5, \ldots,$ $A_k = \frac{2}{k\pi}$, and for $k = 3, 7, \ldots,$ $A_k = -\frac{2}{k\pi}$. Additionally, converting the cosine function to sine function using the identity $cos(\theta) = sin(\theta + \pi/2)$, the θ values become $\theta = \frac{\pi}{2}$. Since A_k alternates by sign every other odd value, the θ value is positive for $k = 1, 5, \ldots$ and negative for $k = 3, 7, \ldots$. Thus, Equation 3 becomes

$$s_2(t) = \sum_{k=1,3,5,\ldots}^{\infty} \frac{2}{k\pi} sin(2\pi kt + \theta) \quad (14)$$

where $\theta = \frac{\pi}{2}$ for $k = 1, 5, \ldots$ and $\theta = -\frac{\pi}{2}$ for $k = 3, 7, \ldots$

1.2 Magnitude and Phase of Discrete-Time Systems

For the following function:

$$y(n) = 0.9y(n-1) + 0.3x(n) + 0.24x(n-1) \quad (15)$$

By setting $x(n) = \delta(n)$, the impulse reponse is found through setting $n = 0$.

$$y(0) = 0.9y(-1) + 0.3\delta(0) + 0.24\delta(-1) = 0.9(0) + 0.3(1) + 0.24(0) = 0.3 \quad (16)$$

$$y(1) = 0.9y(0) + 0.3\delta(1) + 0.24\delta(0) = 0.9(0.3) + 0.3(0) + 0.24(1) = 0.51 \quad (17)$$

For $n > 1$, Equation 15 is just $y(n) = 0.9y(n-1)$. Therefore the impulse is

$$h(n) = 0.3\delta(n) + 0.51(0.9)^{n-1}u(n-1) \quad (18)$$

The system diagram of Equation 15 is shown in Figure 3.

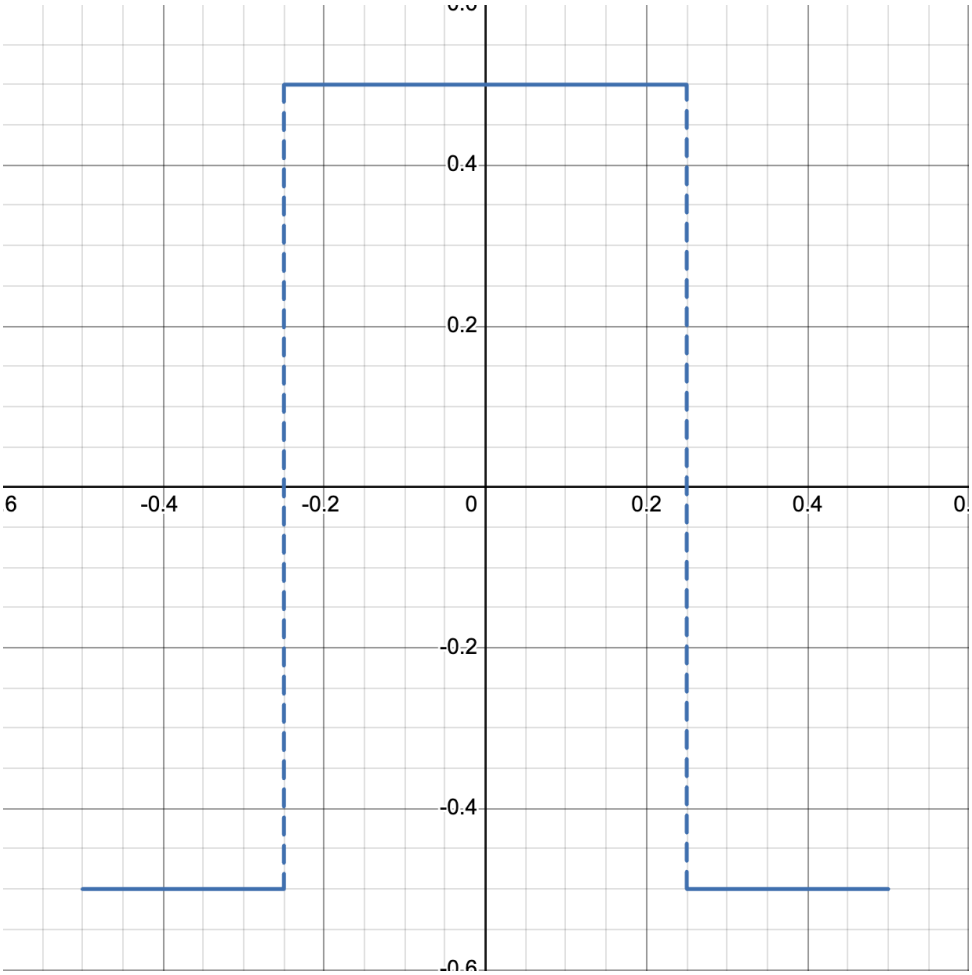

Figure 2: Sketch of Equation 2 from $[-\frac{T_0}{2}, \frac{T_0}{2}]$

The Z-transform of Equation 15 is found using the equation $x(n-k) \leftrightarrow z^{-k}X(z)$

$$Y(z) = 0.9z^{-1}Y(z) + 0.3X(z) + 0.24z^{-1}X(z) \tag{19}$$

The transfer function of Equation 15 using Equation 19 is

$$(1 - 0.9z^{-1})Y(z) = (0.3 + 0.24z^{-1})X(z) \tag{20}$$

$$H(z) = \frac{0.3 + 0.24z^{-1}}{1 - 0.9z^{-1}} \tag{21}$$

The magnitude and phase responses of Equation 21 are plotted in Figure 4

2 Continuous-Time Frequency Analysis

2.1 Synthesis of Periodic Signals

2.2 Modulation Property

Changing the modulating frequency results in a frequency shift from 10Hz to 15Hz as shown in Figures 8*****. The spectrum has a comb structure because the signal being modulated, triangle pulse, is periodic. Periodic signals can be represented by impulses, and the distance between the impulses is found through $f = \frac{1}{T}$. If the period of the triangle pulse were to increase toward infinity, the triangle pulse would no longer have a period. Therefore, the distances between the impulses become small, and the function theoretically remains continuous.

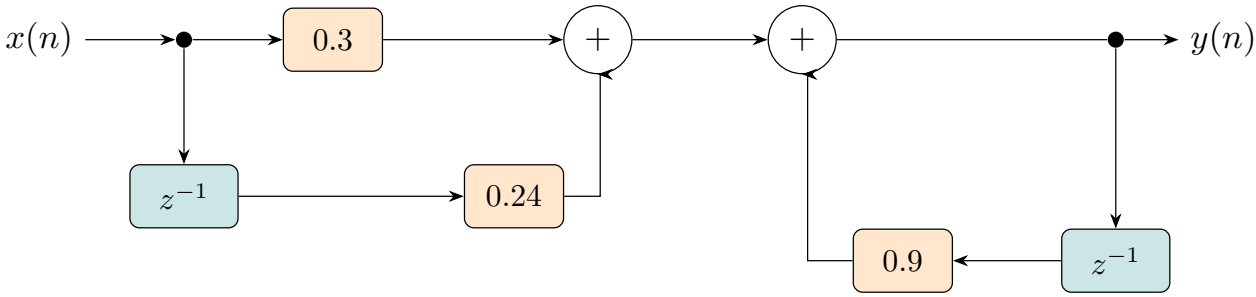

Figure 3: Direct Form I: $y(n) = 0.9\,y(n-1) + 0.3\,x(n) + 0.24\,x(n-1)$

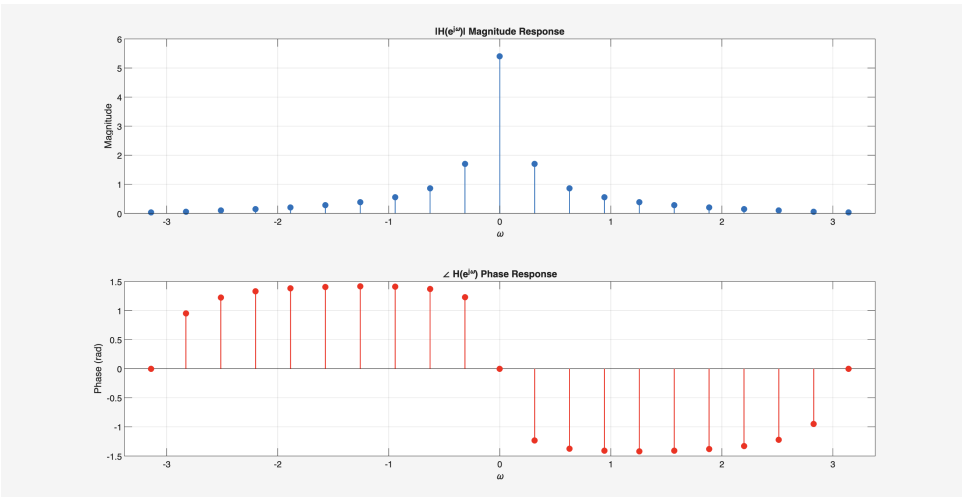

Figure 4: Discrete-Time Magnitude and Phase Responses of Equation 21

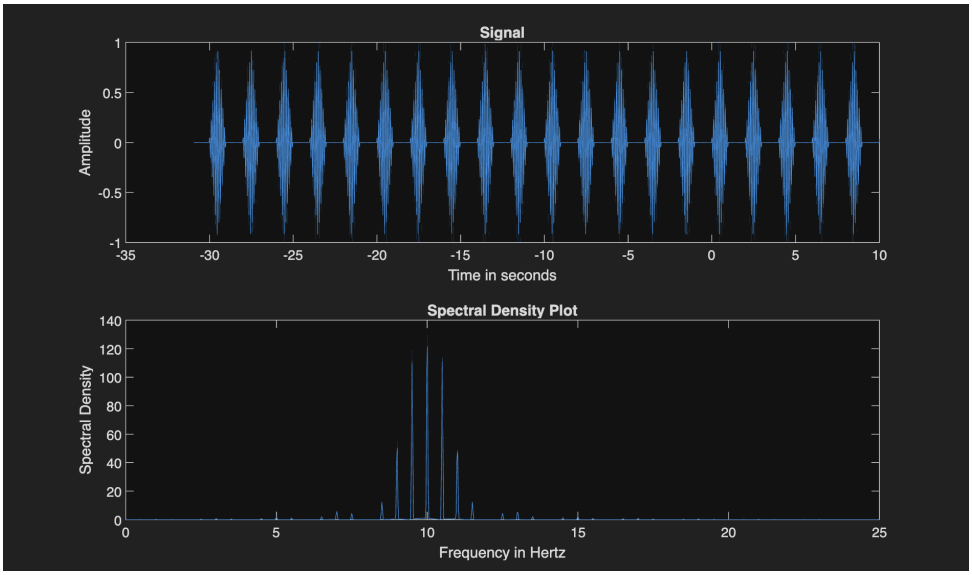

Figure 5: Spectrum Analyzer of 2 sec Triangle Pulse at 10 Hz

2.3 System Analysis

3 Discrete-Time Frequency Analysis

3.1 Discrete-Time Fourier Transform

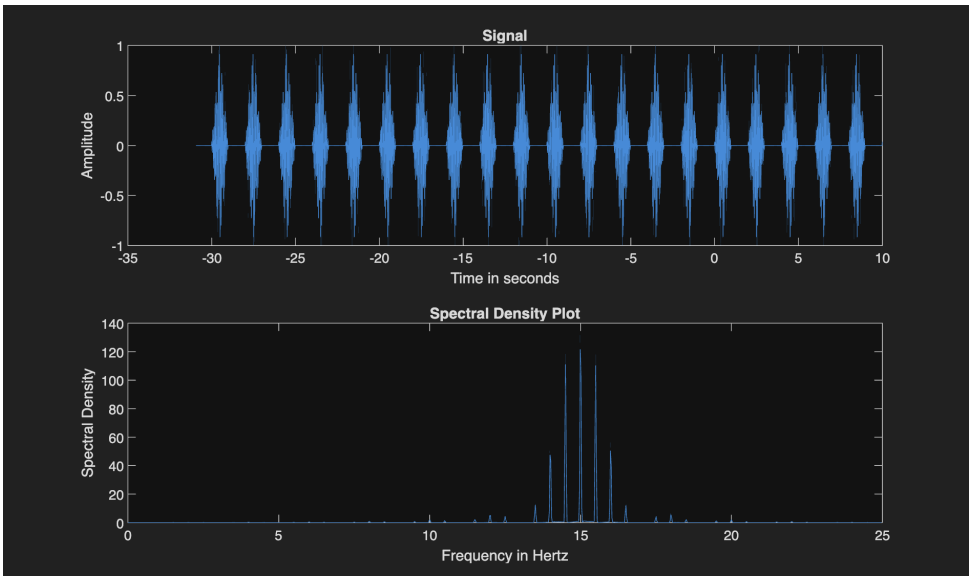

Figure 6: Spectrum Analyzer of 2 sec Triangle Pulse at 15 Hz

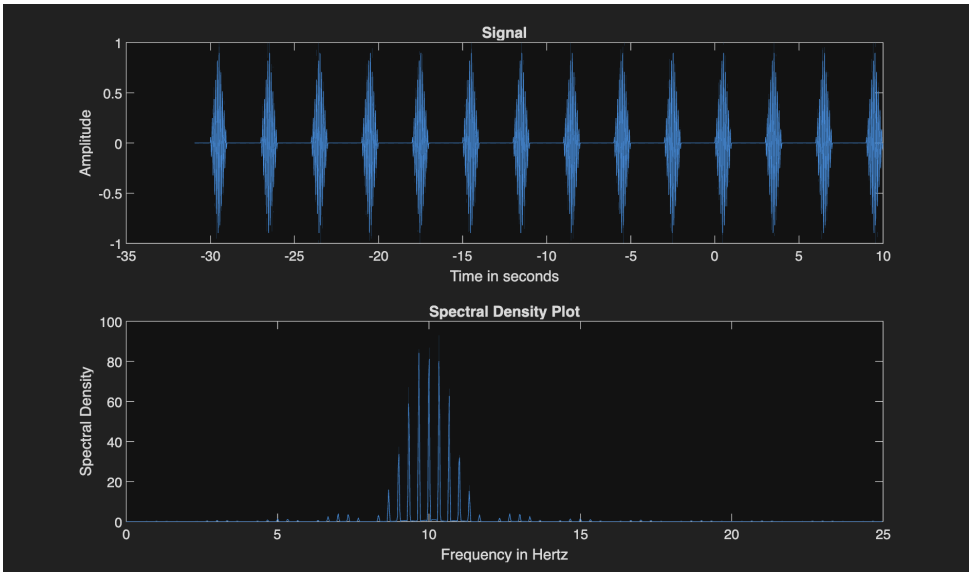

Figure 7: Spectrum Analyzer of 3 sec Triangle Pulse at 10 Hz

dtft.m

```
% Purpose: Computes the Discrete-Time Fourier Transform (DTFT) of a Signal
% Input: (x) - vector representing the time-domain
%        (n0) - time index corresponding to 1st element in vector x
%        (dw) - spacing between samples
% Output: (X) - Values corresponding to DTFT of signal from -pi to pi

function X = dtft(x, n0, dw)
    w = -pi:dw:pi;
    N = length(x);
    n = n0:(n0 + N - 1);
    X = x * exp(-1j * n' * w);
end
```

3.2 System Analysis

The complete block diagram is in Figure 8********. Figures 9, 10, and 11 represent the sine wave at different frequencies. Figure 12 represents the table with the theoretical values.

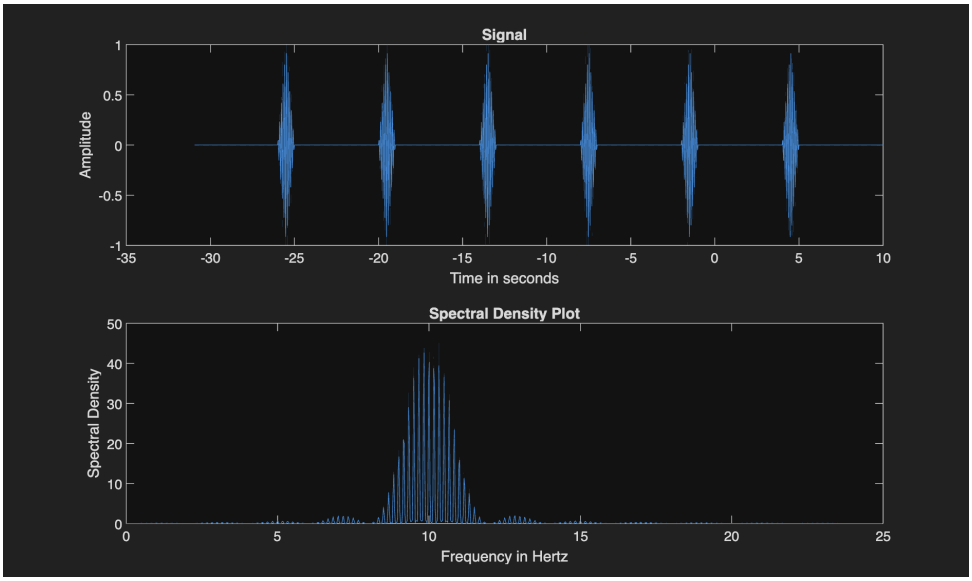

Figure 8: Spectrum Analyzer of 6 sec Triangle Pulse at 10 Hz

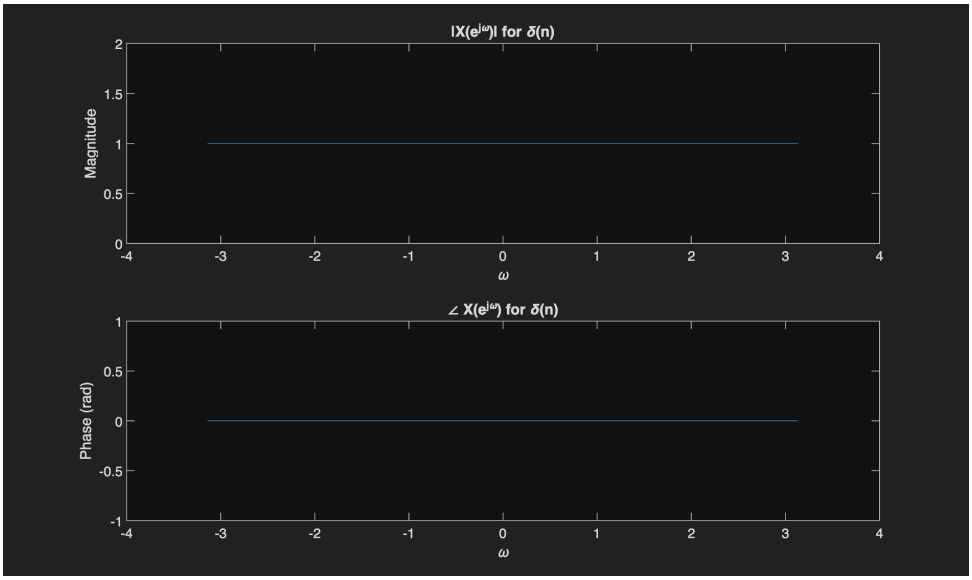

Figure 9: Magnitude and Phase Responses of $X(e^{jw})$

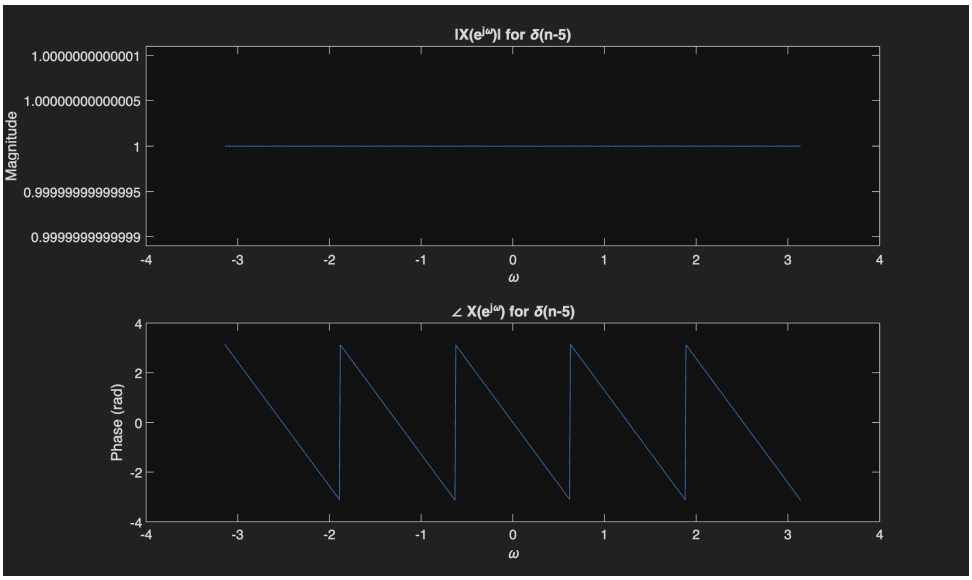

Figure 10: Magnitude and Phase Responses of $X(e^{jw})$

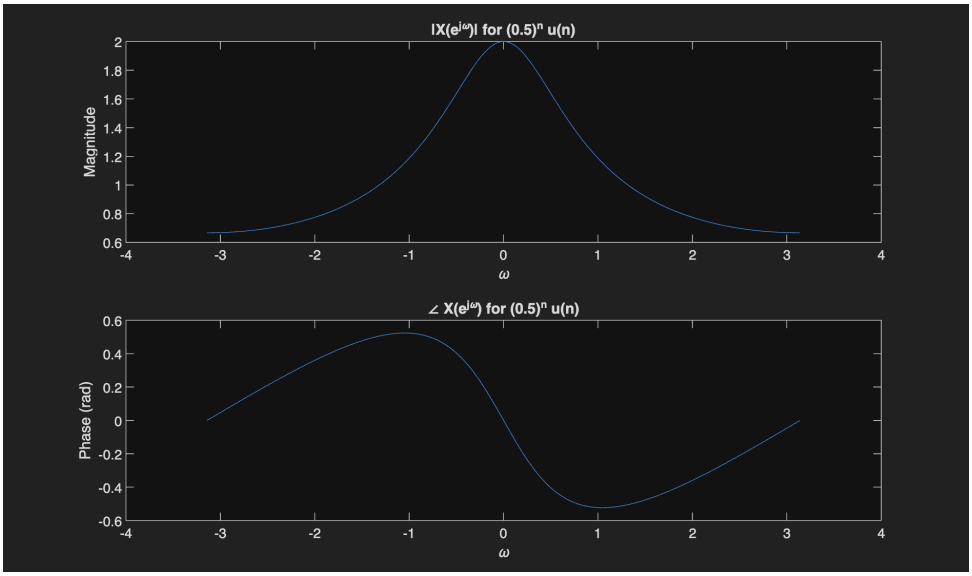

Figure 11: Magnitude and Phase Responses of $X(e^{jw})$

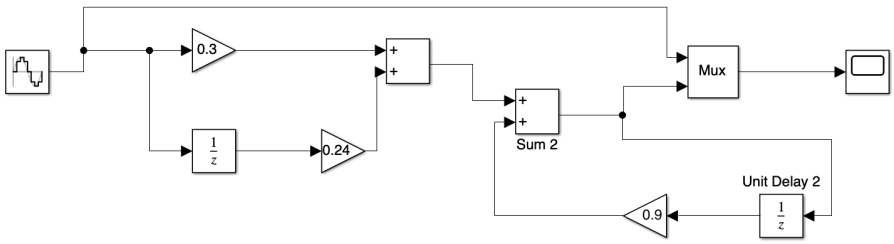

Figure 12: Completed Block Diagram of Discrete-Time System Analysis

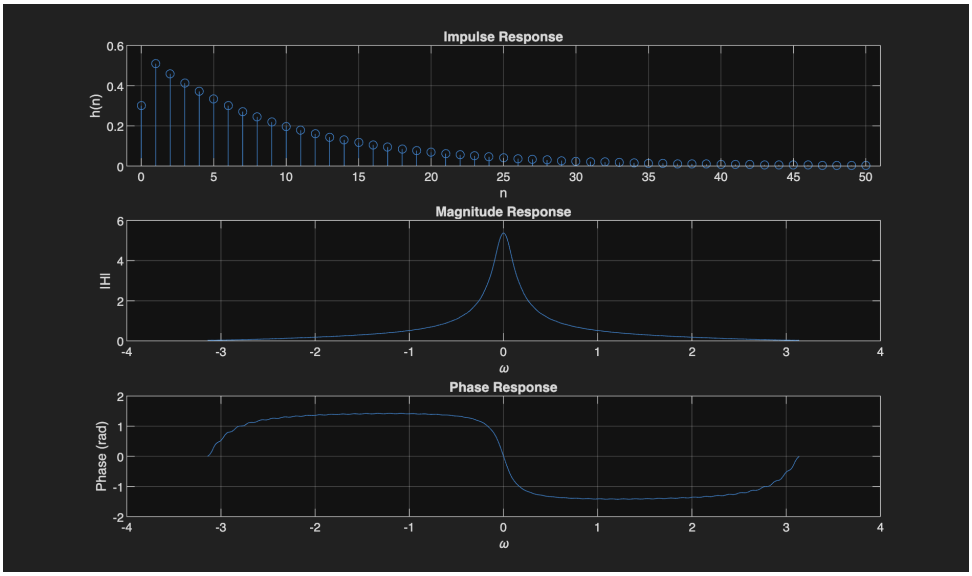

Figure 13: Impulse Response, Magnitude, and Phase of the Frequency Response

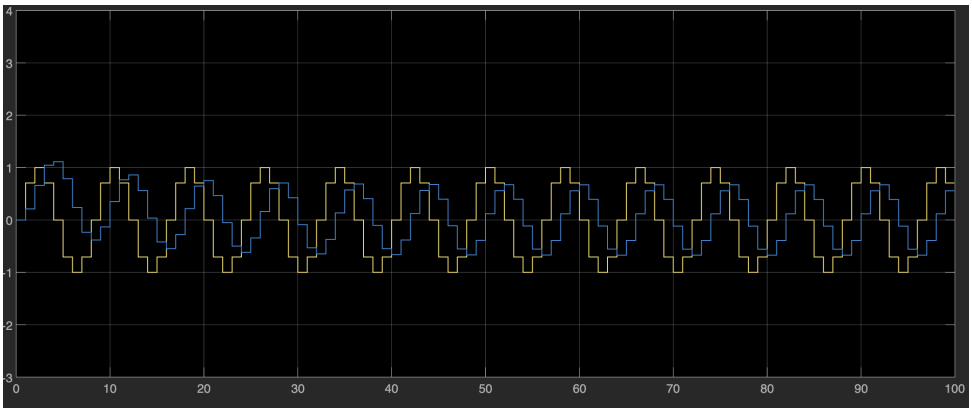

Figure 14: Sine Wave at $w = \pi/4$

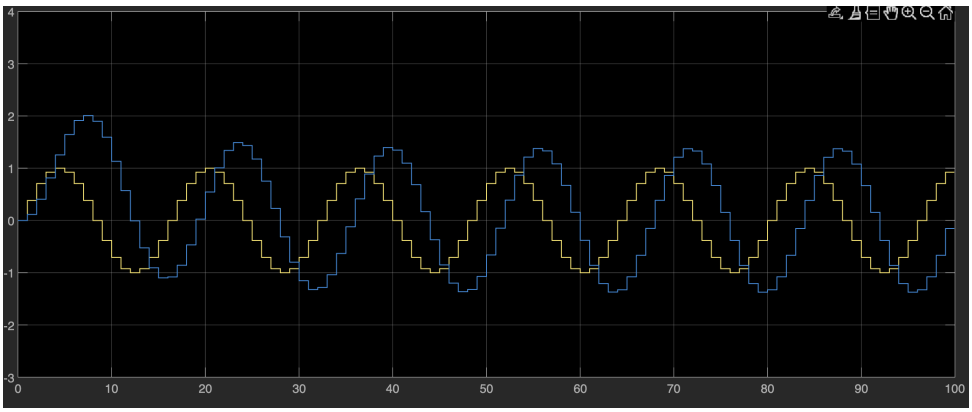

Figure 15: Sine Wave at $w = \pi/8$

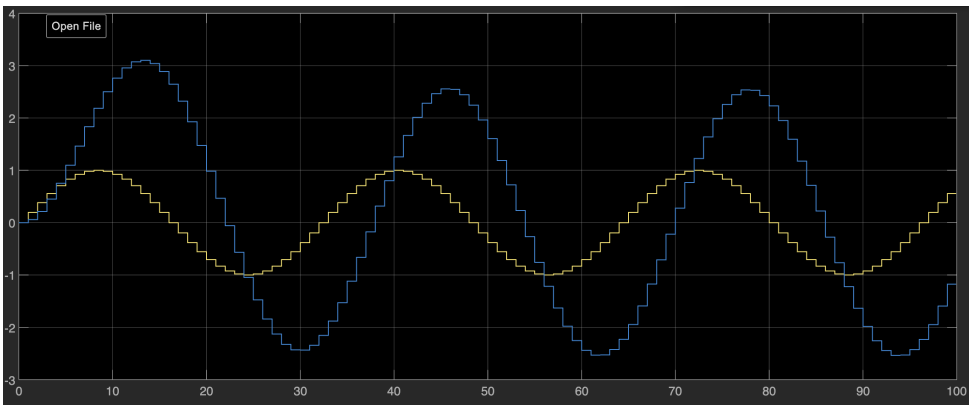

Figure 16: Sine Wave at $w = \pi/16$

Frequency	Theoretical |H|
pi/16	2.5452
pi/8	1.3816
pi/4	0.6814

Figure 17: Table of Theoretical Values with Frequencies